

Fourier Gain and Phase-Error Characterization of a Local LABFM Interpolation Operator

Abstract

A single Fourier mode is used to characterize the local amplitude and phase distortions introduced by a LABFM interpolation operator. Three mathematically equivalent representations are developed. First, the complex amplification function is derived entirely in dimensional coordinates and dimensional wavenumbers. Second, the spatial coordinates and wavenumbers are nondimensionalized consistently, with an additional Nyquist-normalized wavenumber introduced for spectral plotting. Third, only the spatial coordinates are scaled while the physical wavenumber components are retained. Throughout the primary analysis, the departure displacement and Fourier-wave orientation are prescribed, and the full-field extrema are taken only over the stencil centers in the computational domain. The stencil label ℓ uniquely identifies a local stencil model and is not required to equal its cardinality.

1 Dimensional Fourier formulation

1.1 Geometric and stencil notation

Let $\Omega \subset \mathbb{R}^2$ be the computational domain and let

$$\mathcal{I}_\Omega = \{i : \mathbf{x}_i \in \Omega\} \quad (1.1)$$

be the set of admissible stencil-center indices. The index i denotes a stencil center and j denotes a node belonging to the corresponding local stencil. Their positions are

$$\mathbf{x}_i = (x_i, y_i)^\top, \quad \mathbf{x}_j = (x_j, y_j)^\top. \quad (1.2)$$

The relative coordinates of node j with respect to center i are

$$x_{ji} = x_j - x_i, \quad y_{ji} = y_j - y_i. \quad (1.3)$$

The departure point used in the primary analysis is prescribed rather than scanned. Its position relative to center i is

$$\boldsymbol{\delta}_d = \mathbf{x}_d - \mathbf{x}_i = \begin{bmatrix} x_{di} \\ y_{di} \end{bmatrix}, \quad x_{di} = x_d - x_i, \quad y_{di} = y_d - y_i. \quad (1.4)$$

The symbol d therefore identifies the fixed departure point used in the analysis; it is not an index over which a full-field extremum is taken.

For each center i , the set $\mathcal{S}_{i\ell}$ denotes one complete local stencil and contains the center itself,

$$i \in \mathcal{S}_{i\ell}. \quad (1.5)$$

The symbol ℓ is a unique stencil-model label. It may be numerical, for example

$$\ell \in \{10, 15, 17, 31, \dots\}, \quad (1.6)$$

or symbolic, for example $\ell \in \{a, b, c, d, \dots\}$. When a numerical label is used, it commonly coincides with the total number of stencil nodes, including the center, but no algebraic step below requires the identity $|\mathcal{S}_{i\ell}| = \ell$. The only required property is that the pair (i, ℓ) uniquely identifies one local interpolation model.

The local interpolation of a scalar field f at the prescribed departure displacement is written in difference form as

$$\tilde{f}_{i\ell}(\boldsymbol{\delta}_d) = f(\mathbf{x}_i) + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} [f(\mathbf{x}_j) - f(\mathbf{x}_i)]. \quad (1.7)$$

Because $i \in \mathcal{S}_{i\ell}$, the summand corresponding to $j = i$ vanishes identically. The center weight may therefore be set to $w_{ii,d}^{(\ell)} = 0$ without loss of generality, and a separate stencil set excluding the center is unnecessary.

1.2 Single Fourier mode and discrete response

Let k_n and k_m be the dimensional wavenumber components in the x - and y -directions, respectively. The continuous Fourier probing function is

$$f^F(k_n, k_m; \mathbf{x}) = \exp[i(k_n x + k_m y)], \quad \mathbf{x} = (x, y)^\top, \quad i^2 = -1. \quad (1.8)$$

The superscript F identifies the Fourier mode. Since f^F is a continuous function, a spatial location is supplied as a function argument rather than as a discrete subscript.

At node j and at the departure point,

$$f^F(k_n, k_m; \mathbf{x}_j) = f^F(k_n, k_m; \mathbf{x}_i) \exp[i(k_n x_{ji} + k_m y_{ji})], \quad (1.9a)$$

$$f^F(k_n, k_m; \mathbf{x}_i + \boldsymbol{\delta}_d) = f^F(k_n, k_m; \mathbf{x}_i) \exp[i(k_n x_{di} + k_m y_{di})]. \quad (1.9b)$$

Substitution of equation (1.8) into equation (1.7) gives the discrete Fourier response of the local model (i, ℓ) :

$$\begin{aligned} \tilde{f}_{i\ell}^F(k_n, k_m; \boldsymbol{\delta}_d) &= f^F(k_n, k_m; \mathbf{x}_i) + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} [f^F(k_n, k_m; \mathbf{x}_j) - f^F(k_n, k_m; \mathbf{x}_i)] \\ &= f^F(k_n, k_m; \mathbf{x}_i) \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \{ \exp[i(k_n x_{ji} + k_m y_{ji})] - 1 \} \right]. \end{aligned} \quad (1.10)$$

Only the discrete approximation carries the subscript pair (i, ℓ) . The wavenumber components and the departure displacement remain function arguments.

1.3 Complex amplification function, gain factor, and phase error

The local complex amplification function is defined as the ratio between the discrete Fourier response and the exact continuous Fourier value at the same departure point:

$$G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d) = \frac{\tilde{f}_{i\ell}^F(k_n, k_m; \boldsymbol{\delta}_d)}{f^F(k_n, k_m; \mathbf{x}_i + \boldsymbol{\delta}_d)}. \quad (1.11)$$

Using equations (1.10) and (1.9b),

$$\begin{aligned} G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d) &= \exp[-i(k_n x_{di} + k_m y_{di})] \\ &\quad \times \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \{ \exp[i(k_n x_{ji} + k_m y_{ji})] - 1 \} \right]. \end{aligned} \quad (1.12)$$

No normalization has been introduced in equation (1.12). Every exponential argument is dimensionless because $[k_n] = [k_m] = L^{-1}$ and all coordinate differences have units of length. The arbitrary absolute phase at $\mathbf{x}_i$ has cancelled exactly. An exact interpolation gives

$$G_{i\ell} = 1. \quad (1.13)$$

The local gain factor is the modulus of the complex amplification function:

$$\alpha_{i\ell}^{\text{loc}}(k_n, k_m; \boldsymbol{\delta}_d) = |G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d)|. \quad (1.14)$$

Thus, $\alpha_{i\ell}^{\text{loc}} < 1$ represents amplitude attenuation, $\alpha_{i\ell}^{\text{loc}} = 1$ represents exact amplitude preservation, and $\alpha_{i\ell}^{\text{loc}} > 1$ represents amplitude amplification. The signed gain error is

$$\varepsilon_{\alpha, i\ell}^{\text{loc}} = \alpha_{i\ell}^{\text{loc}} - 1. \quad (1.15)$$

Because the exact departure-point phase has already been divided out in equation (1.11), the local phase error is directly

$$\phi_{i\ell}^{\text{loc}}(k_n, k_m; \boldsymbol{\delta}_d) = \text{Arg}[G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d)]. \quad (1.16)$$

The exact phase error is zero. A nonzero value of equation (1.16) represents interpolation-induced numerical dispersion. The phase is undefined when $G_{i\ell} = 0$. Continuous signed phase curves should be unwrapped in wavenumber, whereas a worst-case absolute phase metric may use the principal argument in $(-\pi, \pi]$ [1].

For completeness, the normalized complex interpolation error is

$$\varepsilon_{i\ell}^{\text{loc}} = |1 - G_{i\ell}|, \quad (1.17)$$

and, with $G_{i\ell} = \alpha_{i\ell}^{\text{loc}} \exp(i\phi_{i\ell}^{\text{loc}})$,

$$|1 - G_{i\ell}|^2 = \left(1 - \alpha_{i\ell}^{\text{loc}}\right)^2 + 2\alpha_{i\ell}^{\text{loc}} \left(1 - \cos \phi_{i\ell}^{\text{loc}}\right). \quad (1.18)$$

Hence, the total error combines gain and phase distortions and does not replace their separate examination.

1.4 Fixed wave direction and full-field extrema

Let

$$k = \sqrt{k_n^2 + k_m^2}, \quad k_n = k \cos \theta_k, \quad k_m = k \sin \theta_k. \quad (1.19)$$

The wave orientation θ_k is a curve parameter and is never included in the spatial extremization. Likewise, the departure displacement $\boldsymbol{\delta}_d$ is prescribed and is not scanned. For a fixed pair $(\theta_k, \boldsymbol{\delta}_d)$, the full-field maximum gain is defined only over the stencil centers:

$$\alpha_{\ell}^{\text{max}}(k; \theta_k, \boldsymbol{\delta}_d) = \max_{i \in \mathcal{I}_{\Omega}} |G_{i\ell}(k \cos \theta_k, k \sin \theta_k; \boldsymbol{\delta}_d)|. \quad (1.20)$$

The corresponding maximum absolute phase error is

$$\Phi_{\ell}^{\text{max}}(k; \theta_k, \boldsymbol{\delta}_d) = \max_{i \in \mathcal{I}_{\Omega}} |\text{Arg}[G_{i\ell}(k \cos \theta_k, k \sin \theta_k; \boldsymbol{\delta}_d)]|. \quad (1.21)$$

The strongest amplitude attenuation is characterized by

$$\alpha_{\ell}^{\text{min}}(k; \theta_k, \boldsymbol{\delta}_d) = \min_{i \in \mathcal{I}_{\Omega}} |G_{i\ell}(k \cos \theta_k, k \sin \theta_k; \boldsymbol{\delta}_d)|. \quad (1.22)$$

Thus, $\alpha_\ell^{\max}$ detects the largest local amplification, while $\alpha_\ell^{\min}$ detects the strongest local damping. These quantities are worst-case envelopes of local transfer functions. Because a spatially varying LABFM operator is not generally translation invariant, they should not be interpreted as eigenvalues or as the spectral radius of the assembled global interpolation matrix.

The primary curves considered here use the fixed wave orientation

$$\boxed{\theta_k = \frac{\pi}{4}, \quad k_n = k_m = \frac{k}{\sqrt{2}}.} \quad (1.23)$$

Accordingly,

$$\alpha_\ell^{\max}\left(k; \frac{\pi}{4}, \boldsymbol{\delta}_d\right) = \max_{i \in \mathcal{I}_\Omega} \left| G_{i\ell}\left(\frac{k}{\sqrt{2}}, \frac{k}{\sqrt{2}}; \boldsymbol{\delta}_d\right) \right|, \quad (1.24)$$

with analogous specializations for equations (1.21) and (1.22). Other values of θ_k may be examined subsequently as separate directional studies.

2 Paired nondimensionalization and Nyquist normalization

2.1 Scaled coordinates and phase-scaled wavenumbers

Let A_i be the area associated with center i , and define the local characteristic length

$$s_i = \sqrt{A_i}. \quad (2.1)$$

The dimensionless relative coordinates are

$$\xi_{ji} = \frac{x_{ji}}{s_i}, \quad \eta_{ji} = \frac{y_{ji}}{s_i}, \quad (2.2a)$$

$$\xi_{di} = \frac{x_{di}}{s_i}, \quad \eta_{di} = \frac{y_{di}}{s_i}. \quad (2.2b)$$

The scaled departure vector is

$$\hat{\boldsymbol{\delta}}_{d,i} = \frac{\boldsymbol{\delta}_d}{s_i} = \begin{bmatrix} \xi_{di} \\ \eta_{di} \end{bmatrix}. \quad (2.3)$$

The dimensionless variables that arise directly in the Fourier phase are

$$\mu_{n,i} = k_n s_i, \quad \mu_{m,i} = k_m s_i. \quad (2.4)$$

They preserve the phase exactly:

$$k_n x_{ji} + k_m y_{ji} = \mu_{n,i} \xi_{ji} + \mu_{m,i} \eta_{ji}, \quad (2.5)$$

$$k_n x_{di} + k_m y_{di} = \mu_{n,i} \xi_{di} + \mu_{m,i} \eta_{di}. \quad (2.6)$$

The variables $\mu_{n,i}$ and $\mu_{m,i}$ are dimensionless phase-scaled wavenumbers. They are not yet normalized so that unity represents a Nyquist limit.

2.2 Nominal Nyquist-normalized wavenumbers

For spectral plotting, introduce the nominal local Nyquist reference

$$k_{\text{Ny},i}^{\text{nom}} = \frac{\pi}{s_i}. \quad (2.7)$$

The Nyquist-normalized wavenumber components are then

$$\kappa_{n,i} = \frac{k_n}{k_{\text{Ny},i}^{\text{nom}}} = \frac{k_n s_i}{\pi}, \quad \kappa_{m,i} = \frac{k_m}{k_{\text{Ny},i}^{\text{nom}}} = \frac{k_m s_i}{\pi}. \quad (2.8)$$

Thus,

$$\mu_{n,i} = \pi \kappa_{n,i}, \quad \mu_{m,i} = \pi \kappa_{m,i}. \quad (2.9)$$

The radial Nyquist-normalized wavenumber is

$$\kappa_i = \sqrt{\kappa_{n,i}^2 + \kappa_{m,i}^2} = \frac{k}{k_{\text{Ny},i}^{\text{nom}}} = \frac{k s_i}{\pi}. \quad (2.10)$$

For a fixed orientation,

$$\kappa_{n,i} = \kappa_i \cos \theta_k, \quad \kappa_{m,i} = \kappa_i \sin \theta_k. \quad (2.11)$$

The definition in equation (2.8) is the appropriate normalization when the horizontal axis is intended to represent a fraction of the local Nyquist scale. By contrast, $k_n s_i$ and $k_m s_i$ are the natural phase variables. Both conventions are correct, and they differ only by the factor π . On an irregular point cloud, $k_{\text{Ny},i}^{\text{nom}} = \pi/s_i$ is a nominal local reference rather than a rigorous direction-independent aliasing boundary.

2.3 Amplification function in Nyquist-normalized form

Using equations (2.2) and (2.8), the same amplification function is written as

$$\begin{aligned} G_{i\ell}^{\text{Ny}}(\kappa_{n,i}, \kappa_{m,i}; \hat{\boldsymbol{\delta}}_{d,i}) &= \exp[-i\pi (\kappa_{n,i} \xi_{di} + \kappa_{m,i} \eta_{di})] \\ &\times \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \{ \exp[i\pi (\kappa_{n,i} \xi_{ji} + \kappa_{m,i} \eta_{ji})] - 1 \} \right]. \end{aligned} \quad (2.12)$$

The factor π is essential. It follows directly from the definition $k_{\text{Ny},i}^{\text{nom}} = \pi/s_i$ and preserves the original dimensional Fourier phase. For corresponding dimensional and normalized variables,

$$G_{i\ell}^{\text{Ny}}\left(\frac{k_n s_i}{\pi}, \frac{k_m s_i}{\pi}; \frac{\boldsymbol{\delta}_d}{s_i}\right) = G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d). \quad (2.13)$$

Consequently, the gain factor and phase error are unchanged by this reparameterization.

For full-field resolving-power curves, a common normalized radial parameter κ may be prescribed at every center. In that case, the physical wavenumber used at center i is

$$k^{(i)} = \kappa k_{\text{Ny},i}^{\text{nom}} = \frac{\pi \kappa}{s_i}. \quad (2.14)$$

At the primary angle $\theta_k = \pi/4$,

$$\kappa_n = \kappa_m = \frac{\kappa}{\sqrt{2}}. \quad (2.15)$$

The corresponding full-field maximum gain is

$$\alpha_{\ell, \text{Ny}}^{\text{max}}\left(\kappa; \frac{\pi}{4}, \boldsymbol{\delta}_d\right) = \max_{i \in \mathcal{I}_\Omega} \left| G_{i\ell}^{\text{Ny}}\left(\frac{\kappa}{\sqrt{2}}, \frac{\kappa}{\sqrt{2}}; \hat{\boldsymbol{\delta}}_{d,i}\right) \right|. \quad (2.16)$$

The maximum phase error and minimum gain are defined analogously by replacing the modulus in equation (2.16) with the corresponding operations. A common κ compares all centers at the same wavelength relative to their own local spacing; it does not impose one common physical wavenumber when s_i varies spatially.

3 Coordinate-only scaling with physical wavenumbers

The preceding section scaled both the coordinates and the wavenumbers. An alternative representation scales only the spatial coordinates while retaining the dimensional physical wavenumber components k_n and k_m .

Using

$$x_{ji} = s_i \xi_{ji}, \quad y_{ji} = s_i \eta_{ji}, \quad x_{di} = s_i \xi_{di}, \quad y_{di} = s_i \eta_{di}, \quad (3.1)$$

the Fourier phases become

$$k_n x_{ji} + k_m y_{ji} = s_i (k_n \xi_{ji} + k_m \eta_{ji}), \quad (3.2)$$

$$k_n x_{di} + k_m y_{di} = s_i (k_n \xi_{di} + k_m \eta_{di}). \quad (3.3)$$

The factor s_i must remain explicitly in the phase because the coordinates ξ and η are dimensionless whereas k_n and k_m retain units of inverse length.

The complex amplification function can therefore be written as

$$\boxed{G_{i\ell}^{\text{coord}}(k_n, k_m; \widehat{\boldsymbol{\delta}}_{d,i}, s_i) = \exp[-is_i (k_n \xi_{di} + k_m \eta_{di})] \times \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \{ \exp[is_i (k_n \xi_{ji} + k_m \eta_{ji})] - 1 \} \right]}. \quad (3.4)$$

Equation (3.4) is not a new spectral model. It is an exact coordinate transformation of equation (1.12):

$$\boxed{G_{i\ell}^{\text{coord}}(k_n, k_m; \boldsymbol{\delta}_d / s_i, s_i) = G_{i\ell}(k_n, k_m; \boldsymbol{\delta}_d)}. \quad (3.5)$$

Therefore,

$$|G_{i\ell}^{\text{coord}}| = |G_{i\ell}|, \quad \text{Arg}(G_{i\ell}^{\text{coord}}) = \text{Arg}(G_{i\ell}) \pmod{2\pi}. \quad (3.6)$$

No additional normalization of the gain factor or phase error is required.

At a fixed wave orientation,

$$G_{i\ell}^{\text{coord}}(k; \theta_k, \widehat{\boldsymbol{\delta}}_{d,i}, s_i) = \exp[-ik s_i (\xi_{di} \cos \theta_k + \eta_{di} \sin \theta_k)] \times \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \{ \exp[ik s_i (\xi_{ji} \cos \theta_k + \eta_{ji} \sin \theta_k)] - 1 \} \right]. \quad (3.7)$$

For the primary angle $\theta_k = \pi/4$,

$$\boxed{G_{i\ell}^{\text{coord}}\left(k; \frac{\pi}{4}, \widehat{\boldsymbol{\delta}}_{d,i}, s_i\right) = \exp\left[-i \frac{k s_i}{\sqrt{2}} (\xi_{di} + \eta_{di})\right] \times \left[1 + \sum_{j \in \mathcal{S}_{i\ell}} w_{ji,d}^{(\ell)} \left\{ \exp\left[i \frac{k s_i}{\sqrt{2}} (\xi_{ji} + \eta_{ji})\right] - 1 \right\} \right]}. \quad (3.8)$$

The full-field maximum gain at a common physical wavenumber k remains a maximum over i only:

$$\boxed{\alpha_{\ell, \text{coord}}^{\max}\left(k; \frac{\pi}{4}, \boldsymbol{\delta}_d\right) = \max_{i \in \mathcal{I}_\Omega} \left| G_{i\ell}^{\text{coord}}\left(k; \frac{\pi}{4}, \widehat{\boldsymbol{\delta}}_{d,i}, s_i\right) \right|}. \quad (3.9)$$

Unlike equation (2.16), equation (3.9) imposes the same physical Fourier mode at every center. When s_i varies, the local Nyquist ratio $k s_i / \pi$ consequently varies across the domain.

The coordinate-only form should be interpreted as a rewriting of an already constructed interpolation operator. If the interpolation weights are recomputed after coordinate scaling, the weight-construction procedure must itself be scale consistent for equation (3.5) to remain valid.

4 Recommended interpretation and plotting variables

The three representations answer different practical needs but describe the same local interpolation response:

- (i) The dimensional form, equation (1.12), is the primary mathematical definition and contains no arbitrary normalization.
- (ii) The paired nondimensional form, equation (2.12), is most convenient for comparing resolving power. The horizontal variable $\kappa = k/k_{\text{Ny},i}^{\text{nom}}$ directly indicates the fraction of the nominal local Nyquist scale, and the Fourier exponent contains the required factor π .
- (iii) The coordinate-only form, equation (3.4), retains the physical wavenumbers and merely rewrites the local geometry in units of s_i . It is the appropriate representation when one common physical Fourier mode is imposed throughout a nonuniform domain.

For the present primary analysis, the departure displacement δ_d and the wave orientation $\theta_k = \pi/4$ are fixed. The only full-field extremization variable is the stencil-center index $i \in \mathcal{I}_\Omega$. Curves for other wave orientations should be generated separately rather than by maximizing over θ_k .

The minimum set of reported curves for each stencil model ℓ is therefore

$$\alpha_\ell^{\max}, \quad \alpha_\ell^{\min}, \quad \Phi_\ell^{\max}, \quad (4.1)$$

plotted either against the physical wavenumber k or against the nominal Nyquist-normalized wavenumber κ , with the selected convention stated explicitly.

Elementary consistency checks

The following properties should hold before any spectral curve is interpreted:

$$G_{i\ell}(0, 0; \delta_d) = 1, \quad (4.2)$$

$$G_{i\ell}(-k_n, -k_m; \delta_d) = G_{i\ell}(k_n, k_m; \delta_d)^* \quad \text{for real weights}, \quad (4.3)$$

$$|G_{i\ell}| \rightarrow 1, \quad \text{Arg}(G_{i\ell}) \rightarrow 0 \quad \text{as } k \rightarrow 0. \quad (4.4)$$

The first identity follows exactly from the difference form in equation (1.7); the center-inclusive definition of $\mathcal{S}_{i\ell}$ does not alter it because the $j = i$ contribution is zero.

References

- [1] D. Desvigne, O. Marsden, C. Bogey, and C. Bailly, “Development of noncentered wavenumber-based optimized interpolation schemes with amplification control for overlapping grids,” *SIAM Journal on Scientific Computing*, vol. 32, no. 4, pp. 2074–2098, 2010. doi:10.1137/090758702.